

ANGULARJS

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ANGULARJS CONTROLLER

- The controller in AngularJS is a JavaScript function that maintains the application data and behavior using **\$scope** object.
- Attach properties and methods to the **\$scope** object

Example: AngularJS Controller

```
<!DOCTYPE html>
<html >
<head>
  <title>AngularJS Controller</title>
  <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="myController">
    {{message}}
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('myController', function ($scope) {
      $scope.message = "Hello World!";
    });
  </script>
</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="myController">
    {{message}}
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);
    ngApp.controller('myController', function ($scope) {
      $scope.message = "Hello World!";
    });
  </script>
</body>
</html>
```

1. Specify a controller using ng-controller

2. Create an App module

3. Create a Controller

5. Use a property created inside a controller

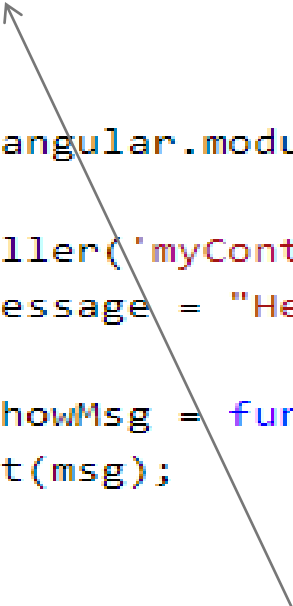
4. Attach a property to \$scope

ATTACH BEHAVIORS

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src="/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="myController">
    Enter Message: <input type="text" ng-model="message" /> <br />
    <button ng-click="showMsg(message)">Show Message</button>
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('myController', function ($scope) {
      $scope.message = "Hello World!";

      $scope.showMsg = function (msg) {
        alert(msg);
      };
    });
  </script>
</body>
</html>
```



The **ng-click** directive is used to handle click event in AngularJS application.

Result:

Enter Message:

Show Message

Hello World!

OK

webx.0

OK

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src=~/Scripts/angular.js></script>
</head>
<body ng-app="myNgApp">
  <div id="div1" ng-controller="myController">
    Message: {{message}} <br />
    <div id="div2">
      Message: {{message}}
    </div>
  </div>
  <div id="div3">
    Message: {{message}}
  </div>
  <div id="div4" ng-controller="anotherController">
    Message: {{message}}
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('myController', function ($scope) {
      $scope.message = "This is myController";
    });

    ngApp.controller('anotherController', function ($scope) {
      $scope.message = "This is anotherController";
    });
  </script>
</body>
```

Result:

Message: This is myController

Message: This is myController

Message:

Message: This is anotherController

Example: Attach an object

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <h2>Student Information:</h2>
  <div ng-controller="myController">
    First Name: {{student.firstName}} <br />
    Last Name: {{student.lastName}}
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('myController', function ($scope) {
      $scope.student = { firstName: 'James', lastName: 'Bond' };
    });
  </script>
</body>
</html>
```

Result:

First Name: James
Last Name: Bond

Example: Nested Controllers

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src="~/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="parentController">
    Message: {{message1}}
    <div ng-controller="childController">
      Parent Message: {{message1}} </br>
      Child Message: {{message2}}
    </div>
    Child Message: {{message2}}
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('parentController', function ($scope) {
      $scope.message1 = "This is parentController";
    });

    ngApp.controller('childController', function ($scope) {
      $scope.message2 = "This is childController";
    });
  </script>
</body>
</html>
```

Result:

Message: This is parentController

Parent Message: This is parentController

Child Message: This is childController

Child Message:

SCOPE IN ANGULARJS

- The \$scope in an AngularJS is a built-in object, which contains application data and methods.
- The \$scope is glue between a controller and view (HTML). It transfers data from the controller to view and vice-versa.



Example: \$scope

```
<!DOCTYPE html>
<html >
<head>
  <title>AngularJS $scope object</title>
  <script src="/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="myController">
    Message: <br />
    {{message}}<br />
    <span ng-bind="message"></span> <br />
    <input type="text" ng-model="message" />
  </div>
  <script>
    var ngApp = angular.module('myNgApp', []);

    ngApp.controller('myController', function ($scope) {
      $scope.message = "Hello World!";
    });
  </script>
</body>
</html>
```

Diagram illustrating the AngularJS code example with annotations:

- Expression**: Points to the `{{message}}` interpolation in the `Message: <br /> {{message}}<br />` line.
- ng-model**: Points to the `ng-model="message"` attribute in the `<input type="text" ng-model="message" />` line.
- ng-bind**: Points to the `ng-bind="message"` attribute in the `<span ng-bind="message"></span>` line.

\$ROOTSCOPE

- An AngularJS application has a single `$rootScope`.
All the other `$scope` objects are child objects.
- The properties and methods attached to `$rootScope` will be available to all the controllers.

Example: \$rootScope & \$scope

```
<!DOCTYPE html>
<html>
<head>
  <title>AngularJS Controller</title>
  <script src="/Scripts/angular.js"></script>
</head>
<body ng-app="myNgApp">
  <div ng-controller="parentController">
    Controller Name: {{controllerName}} <br />
    Message: {{message}} <br />
    <div style="margin:10px 0 10px 20px;" ng-controller="childController">
      Controller Name: {{controllerName}} <br />
      Message: {{message}} <br />
    </div>
  </div>
  <div ng-controller="siblingController">
    Controller Name: {{controllerName}} <br />
    Message: {{message}} <br />
  </div>
```

```
<script>
  var ngApp = angular.module('myNgApp', []);

  ngApp.controller('parentController', function ($scope, $rootScope) {
    $scope.controllerName = "parentController";
    $rootScope.message = "Hello World!";
  });

  ngApp.controller('childController', function ($scope) {
    $scope.controllerName = "childController";
  });

  ngApp.controller('siblingController', function ($scope) {

  });
</script>
</body>
</html>
```


Result:

```
Controller Name: parentController
```

```
Message: Hello World!
```

```
    Controller Name: childController
```

```
    Message: Hello World!
```

```
Controller Name:
```

```
Message: Hello World!
```

ANGULARJS EVENTS

- AngularJS includes certain directives which can be used to provide custom behavior on various DOM events, such as click, dblclick, mouseenter etc.

ANGULARJS EVENT DIRECTIVES

Event Directive
ng-blur
ng-change
ng-click
ng-dblclick
ng-focus
ng-keydown
ng-keyup
ng-keypress
ng-mousedown
ng-mouseenter
ng-mouseleave
ng-mousemove
ng-mouseover
ng-mouseup

Example: Mouse Events

```
<!DOCTYPE html>
<html>
<head>
  <script src="../../Scripts/angular.js"></script>
  <style>
    .redDiv {
      width: 100px;
      height: 100px;
      background-color: red;
      padding: 2px 2px 2px 2px;
    }

    .yellowDiv {
      width: 100px;
      height: 100px;
      background-color: yellow;
      padding: 2px 2px 2px 2px;
    }
  </style>
</head>
```

|
<body ng-app>

<div ng-class="{redDiv: enter, yellowDiv: leave}"

ng-mouseenter="enter=true;leave=false;"

ng-mouseleave="leave=true;enter=false">

Mouse Enter

Leave

</div>

</body>

</html>

Result:

AngularJS Mouse Events Demo:

Mouse

AngularJS Mouse Events Demo:

Mouse Enter



AngularJS Mouse Events Demo:

Mouse Leave



THANK YOU